VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 1 | PAPER - 1

JEE 2022

Date	Maximam Marks	Duration
10th January, 2021	198	3 Hours

General Instructions

1. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).

2. **Section 1** contains **6 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

   Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

   Section 3 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

3. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : _______________

Roll Number : _______________

OMR Bar Code Number : _______________

Candidate's Signature : _______________ Invigilator's Signature _______________

MARKING SCHEME

SECTION - 1 (Maximum Marks: 18)

- This section contains **SIX (06)** Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If only (all) the correct option(s) is(are) chosen
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: –1 In all other cases.

SECTION – 2 (Maximum Marks: 24)

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: –2 In all other cases.

SECTION – 3 (Maximum Marks: 24)

- This section contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

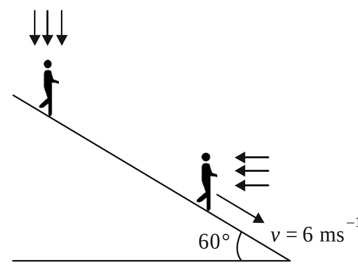
Full Marks: +4 If ONLY the correct Integer value is entered. There is **NO negative marking.**
Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS 66 MARKS

SECTION 1
SINGLE CORRECT ANSWERS TYPE

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A man standing on an inclined plane observes that rain is falling vertically. When he starts moving down the inclined plane with velocity v = 6 m/s, he observes that rain is hitting him horizontally. The angle of inclination of the inclined plane is 60°. The actual speed of rain is:



(A) $2\sqrt{3}\ \text{ms}^{-1}$ **(B)** $3\sqrt{3}\,\text{ms}^{-1}$

(C) $3\ \text{ms}^{-1}$ **(D)** $6\ \text{ms}^{-1}$

2. From the top of a building of height h, a stone A is thrown vertically upwards with velocity v_0. At the same instant, another stone B is thrown vertically downwards with velocity v_0 from the same point. The stone A reaches its highest point at the same instant the stone B reaches the ground. The velocity v_0 is equal to:

(A) $\sqrt{2gh}$ **(B)** $\sqrt{gh}$ **(C)** $\sqrt{\frac{2gh}{3}}$ **(D)** $\sqrt{\frac{gh}{2}}$

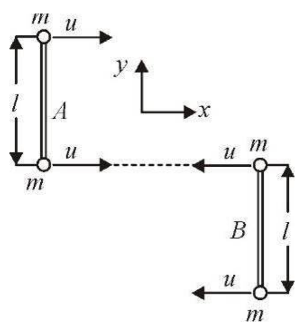
3. Two identical dumbbells are fabricated by welding small identical particles each of mass m at the ends of two identical massless rods each of length l. The dumbbells are moving in free space with equal speeds u towards each other without rotation as shown in the figure. Collisions between the particles are perfectly elastic. Angular speed of each dumbbell after the first collision is:

(A) $\frac{u}{2l}$

(B) $\frac{u}{l}$

(C) $\frac{2u}{l}$

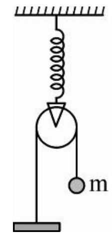
(D) $\frac{4u}{l}$



4. When a glass capillary tube is immersed in a liquid, the liquid rises to a height of 6mm. The tube is gradually pressed down, until a length of only 4mm projects outside. In this situation the liquid meniscus makes an angle of $\theta°$ with the walls of the capillary. If angle of contact for the glass and liquid pair is $\cos^{-1} 3/4$, find θ.

(A) 30° **(B)** 60° **(C)** 37° **(D)** $\cos^{-1}(3/7)$

5. A body of mass m hangs by an inextensible string that passes over a smooth massless pulley that is fitted with a light spring of stiffness k as shown in the figure. If the body is released from rest and the spring is initially relaxed, the maximum elongation of the spring is :



(A) $\frac{4mg}{k}$ **(B)** $\frac{2mg}{k}$

(C) $\frac{mg}{k}$ **(D)** $mg/2k$

6. There is a deep bore well of depth $R/5$ inside earth along radial direction, R being the radius of earth. A small particle is released inside the bore well from the surface of earth so that it falls freely without touching the walls of the well. As it falls into the well, its acceleration (before reaching the bottom of well):

(A) remains constant **(B)** increases continuously

(C) decreasing continuously **(D)** increases then decreases

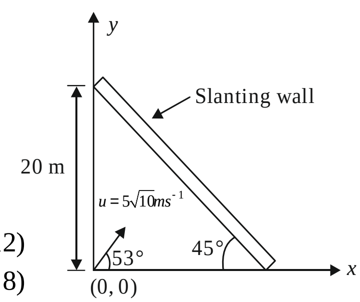
SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. A projectile is fired with a speed of $5\sqrt{10}\ \text{ms}^{-1}$ at an angle 53° with the horizontal towards a fixed, slanting wall. Let the point of projection be the origin of coordinates and let the X and Y axes be along the ground and perpendicular to the ground as shown. Which of these options is/are correct? $(g = 10\ \text{m/s}^2)$



(A) Coordinates of the point where projectile hits the wall are (8, 12)

(B) Coordinates of the point where projectile hits the wall are (12, 8)

(C) Velocity of the projectile when it hits the wall is $(3\sqrt{10}\hat{i} + 1.6\hat{j})$ m/s

(D) Velocity of the projectile when it hits the wall is $3\sqrt{10}\hat{i}$ m/s

8. In an imaginary gravitational field, the gravitational potential energy of a particle of mass m at a distance r from the centre of the field force is given by $u = \frac{kr^2}{2}$, where k is a positive constant of appropriate dimensions. If the particle is moving in a circular orbit of radius r about the centre, choose the correct options.

(A) Total energy of the particle is $-\frac{kr^2}{2}$

(B) Total energy of the particle is kr^2

(C) The orbital time period of the particle is proportional to $r^{3/2}$

(D) The angular momentum of the particle about the centre is proportional to r^2

9. A small steel sphere of density $\frac{31}{4}\times10^3$ kg/m^3 and radius r = 2mm is dropped into a column of castor oil of density $1000\ \text{kg/m}^3$ and coefficient of viscosity $\frac{2}{3}$ kg m$^{-1}s^{-1}$. Which of these options is/are correct? (Take $g = 10\text{m/s}^2$ and $\pi = \frac{25}{8}$)

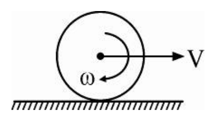
(A) The terminal velocity of the sphere is 9.0 cm/s

(B) The terminal velocity of the sphere is 15.0 cm/s

(C) The rate of heat dissipation due to viscous force after the sphere attains terminal velocity is close to 2×10^{-4} W

(D) The rate of heat dissipation due to viscous force after the sphere attains terminal velocity is close to 2×10^{-5} W

10. A sphere of radius R is moving on a fixed rough surface with linear velocity V and angular velocity ω as shown in figure. Which of these statements is true regarding work done by friction force?



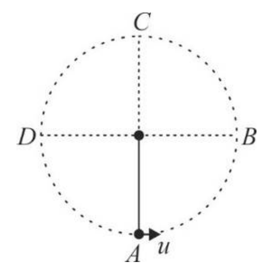
(A) If $V = \omega R$, work done by friction on sphere is zero

(B) If $V > \omega R$, work done by friction on sphere in negative

(C) If $V < \omega R$, work done by friction on sphere is positive

(D) Work done by friction on surface is zero in all cases

11. A small bob of mass m is suspended from a light string of length ℓ whose upper end is tied to a peg. The bob is given a horizontal velocity u at the bottommost point A such that it completes the vertical circle $ABCD$ and tension in the string at the topmost point C is mg. Which of these options are correct?



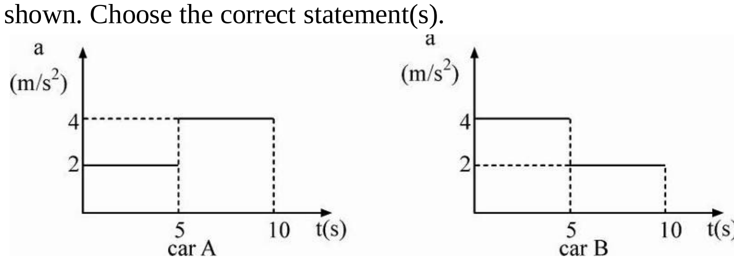
(A) $u = \sqrt{5g\ell}$

(B) $u = \sqrt{6g\ell}$

(C) Tension in the string when the bob passes through point B is 4 mg

(D) As the bob moves from A to C, tension in the string decreases continuously

12. Two cars A and B start from rest and move together for the same time interval. Their acceleration-time graphs are as shown. Choose the correct statement(s).



(A) Both the cars attain the same velocity after 10*s*

(B) The velocity of car B is more than that of car A after 10*s*

(C) The distance covered by car B is more than that of car Aafter 10*s*

(D) Both the cars travel the same distance after 10*s*

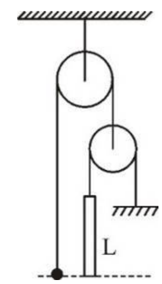
SPACE FOR ROUGH WORK

SECTION 3
NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

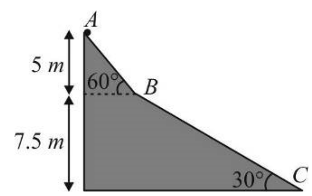
13. When two forces of magnitudes F_1 and F_2 act on a body with an angle $\theta_1 = 60°$ between them, the magnitude of their resultant is R_1. If the angle between the forces is changed to $\theta_2 = 120°$ without changing their magnitudes, the magnitude of their resultant becomes R_2. If the ratio $\frac{R_1}{R_2}$ is equal to $\sqrt{\frac{19}{7}}$, the ratio $\frac{F_1}{F_2}$ is equal to _______. (Given $F_1 > F_2$)

14. In the arrangement shown, the mass of the ball is $\frac{3}{2}$ times the mass of the rod. The length of the rod is L. The masses of the pulleys and threads are negligible. The ball is set on the same horizontal level as the lower end of the rod and then the system is released from rest. The time after which the ball is on the same horizontal level as the upper end of the rod is $\left(X\left(\frac{L}{g}\right)\right)^{1/2}$. The value of X is _________.

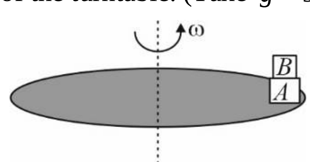


15. A plastic sphere floats in water with 20% of its volume submerged while it floats in oil with 50% of its volume submerged. Find the density of the oil (in kg/m^3). Density of water is $1000\ \text{kg/m}^3$.

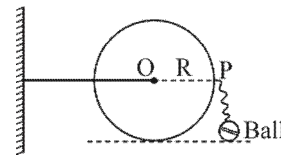
16. A small particle is released from the top of a smooth inclined plane *ABC*. The angle of the incline suddenly changes from 60° to 30° at point *B*. Assume that collision between the particle and the incline is totally inelastic. Find the speed (in m/s) of the particle when it reaches the bottom *C* of the incline. (Take $g = 10\ \text{m/s}^2$)



17. A circular turntable of radius 1m is rotating about its axis with angular velocity ω. Block A of mass 1 kg is placed at the edge of the table and another block B of mass 0.5 kg is placed over A. Coefficient of friction between A and B is 0.2 while that between A and table is 0.45. What can be the maximum value of ω (in rad/s) so that there is no slipping anywhere. Assume that the size of the blocks is very small as complared to the radius of the turntable. (Take $g = 9.8\ \text{m/s}^2$)



18. Figure shows a disc of mass M and radius R hinged at the centre. A small ball of mass $M/2$ is attached to point P with a thread of length $2R$ and held at rest at position shown. Now, the ball is released to fall under gravity. If angular speed with which the disc starts turning when the string becomes taut is $\sqrt{k\dfrac{g}{R}}$, then find k.



SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY 66 MARKS

SECTION 1

SINGLE CORRECT ANSWERS TYPE

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. An organic compound containing carbon, hydrogen and oxygen requires 2.5 L of oxygen to burn completely at STP and gives two litres each of CO_2 and H_2O vapor. The empirical formula of the compound is likely to be:

(A) C_2H_4O **(B)** CH_2O **(C)** C_2H_2O **(D)** C_4H_2O

2. Enthalpy change for the transition of solid Sn, from the grey form (density = 5.75 g cm^{-3}) to the white form (density = 7.31 g cm^{-3}) at 10.0 bar and 298 K is +2.1 kJ mol^{-1}. The internal energy change for this transition in kJ mol^{-1} is (1.0 bar = 10^5 Pa; atomic weight of Sn = 118.7)

(A) –4.4 **(B)** –2.1 **(C)** +2.1 **(D)** +4.4

3. The F–N–F bond angle in NF_3 is 102°30', whereas H–N–H bond angle in NH_3 is 107°48'. This difference in bond angle can be explained based on

(A) VSEPR theory that the repulsion between bond pairs is less in NF_3 than in NH_3

(B) VSEPR theory that the repulsion between bond pairs is more in NF_3 than in NH_3

(C) molecular orbital theory that predicts low bond order for NF_3 molecule

(D) the fact that the first ionization energy of fluorine is higher than that of hydrogen

4. Among 18 groups in the periodic table, the only group that contains examples of elements that are gas, liquid and solid at room temperature (25°C) is:

(A) 1 **(B)** 17 **(C)** 12 **(D)** **6**

5. Among the following, the correct statement(s) regarding $N(SiMe_3)_3$ is/are: [Me is methyl group]

(I) It has a pyramidal geometry around nitrogen

(II) $p\pi - d\pi$ bonding is present in this compound

(III) It is a strong Lewis base

(IV) It has a trigonal planar geometry around nitrogen

(A) (I) and (IV) **(B)** (II) **(C)** (II) and (IV) **(D)** (III)

6. A photon of wavelength 300 nm is absorbed by a gas and then re-emitted as two photons, one with wavelength of 760 nm. The wavelength (in nm) of the second photon is approximately.

(A) 496 **(B)** 300 **(C)** 760 **(D)** 530

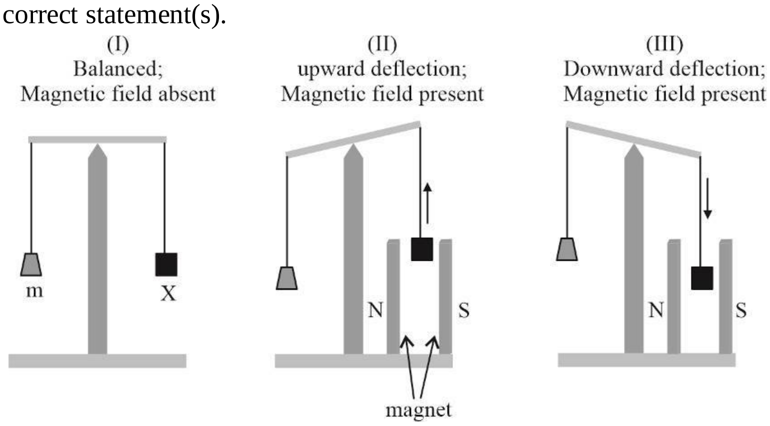
SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. In an experiment, m grams of a compound X (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure I below. In the presence of a magnetic field, the pan with X is either deflected upwards (figure II), or deflected downwards (figure III), depending on the compound X. Identify the correct statement(s).



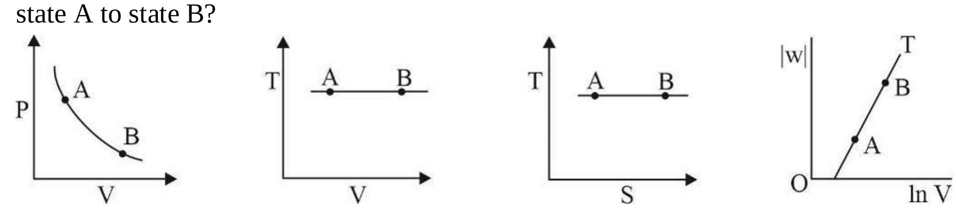
(A) If X is $H_2O(l)$, deflection of the pan is upwards

(B) If X is $Zn(s)$, deflection of the pan is upwards

(C) If X is $O_2\,(g)$, deflection of the pan is downwards

(D) If X is $NO_2(g)$, deflection of the pan is downwards

8. Which of the following diagram is correct for reversible isothermal expansion of an ideal gas from state A to state B?



(A) **(B)** **(C)** **(D)**

Where P = pressure, V = volume, S = entropy, w = work done and T = temperature

9. Which of the following is(are) correct?

(A) HF is more polar than HBr

(B) CuCl is more covalent than NaCl

(C) HF is less polar than HBr

(D) Order of dipole moment is $HF < H_2S < H_2O$

10. Which of the following is(are) correct regarding O, F and Cl?

(A) Fluorine is most electronegative element

(B) Chlorine has maximum electron affinity

(C) Chlorine can expand its octet

(D) HOF is acidic while HOCl is basic

11. Which of the following is/are correct for organic compound given below?

$$CH_2{=}CH{-}CH(CH_3)_2$$

(A) In this compound there is one vinyl group

(B) In this compound there is one isopropyl group

(C) In this compound there are six allylic H atoms

(D) IUPAC name of this compound is 3-Methyl-But-1-ene

12. Which of the following is/are correct for Fe and Fe^{2+}? [Given: Atomic No.: Fe =26]

(A) Both contains equal number of d electrons

(B) Both have same value of spin only magnetic moments

(C) Both have same number of unpaired electrons

(D) Both have same size

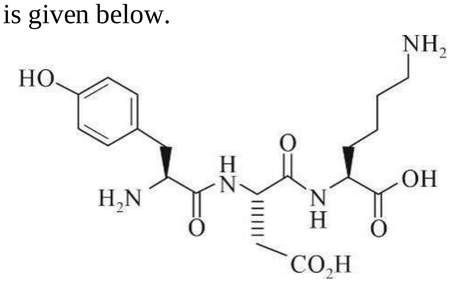
SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

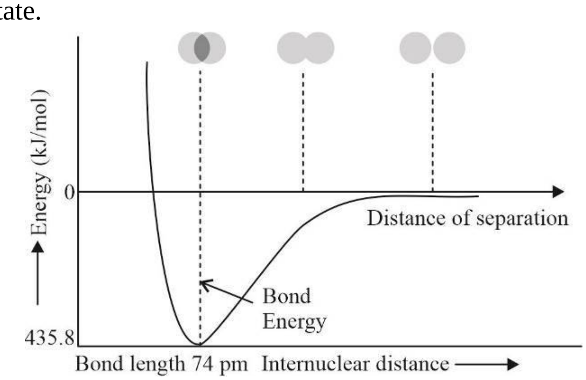
This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

13. The structure of a peptide is given below.



If number of sp^2 hybridized carbon atoms is Z_1, sp^3 hybridized carbon atoms is Z_2 and number of lone pair of electrons is Z_3 respectively, then what is $Z_1+Z_2+Z_3$?

14. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state.



What is the value of energy required in kJ for following gas phase ionization process?

$H_2(g) \rightarrow 2H^+(g) + 2e^-$

Use Avogadro constant as $6.023 \times 10^{23}\ mol^{-1}$ and the energy of the ground state of H atom is $= -2.18 \times 10^{-18}$ J/atom.

15. Tin is obtained from cassiterite (SnO_2) by reduction with coke. Use the data given below to determine the minimum temperature (in K) above which the reduction of cassiterite by coke is spontaneous?

$SnO_2(s) + C(s) \rightarrow Sn(s) + CO_2(g)$

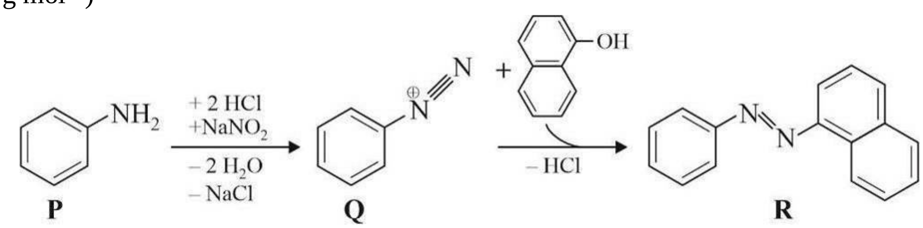
At $298K : \Delta_f H°(SnO_2(s)) = -581.0\,kJ\,mol^{-1}, \Delta_f H°(CO_2(g)) = -394.0\,kJ\,mol^{-1}$,

$S°(SnO_2(s)) = 56.0\,J\,K^{-1}mol^{-1}, S°(Sn(s)) = 52.0\,J\,K^{-1}mol^{-1}$,

$S°(C(s)) = 6.0\,J\,K^{-1}\,mol^{-1}, S°(CO_2(g)) = 210.0\,J\,K^{-1}mol^{-1}$.

Assume that the enthalpies and the entropies are temperature independent.

16. Consider the reaction sequence from **P** to **R** shown below. The yield of the major product **Q** from **P** is 75% and the yield of the major product **R** from **Q** is 50%. What is the amount in grams of **R** obtained from 93 mL of **P**? (Use density of **P** = 1.00 g mL^{-1}; Molar mass of **P** = 93 g mol^{-1}, **Q** = 144 g mol^{-1}, **R** = 248 g mol^{-1})



17. At equilibrium, an acidified solution of 0.05 M Zn^{2+} is saturated with 0.1 M H_2S. What is the minimum molar concentration (in M) of H^+ required to prevent the precipitation of ZnS according to the reaction given below?

$Zn^{2+}(aq) + H_2S(aq) \rightleftharpoons ZnS(s) + 2H^+(aq); Kc = 8$

18. Calculate lattice energy of $MgO_{(s)}$ in kJ/mol if it is given that:

Enthalpy of formation of $MgO_{(s)} = -602\,kJ\,/\,mol$

Enthalpy of sublimation of $Mg_{(s)} = 148\,kJ\,/\,mol$

1st and 2nd ionization energy of Mg = 738 & 1450 kJ/mol respectively

Bond dissociation energy of [O = O] = 498 kJ/mol

1st and 2nd electron gain enthalpy of oxygen = – 141, + 844 kJ/mol respectively.

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS 66 MARKS

SECTION 1
SINGLE CORRECT ANSWERS TYPE

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If the equation of an ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $(a>b)$, where a & b are parameters such that $a^2+b^2=1$, then locus of end points of its latus rectum is:

(A) $(1+x^2)y^2=(1-x^2)^2$ **(B)** $(1+x^2)y^2=2(1-x^2)^2$

(C) $2(1+x^2)y^2=(1-x^2)^2$ **(D)** None of these

2. If z_1 be a complex number such that $|z_1|=13$ & z_2 be complex number such that $|z_2+3-4i|=5$, then number of integers in the range of $|z_1-z_2|$ is:

(A) 20 **(B)** 21 **(C)** 22 **(D)** 23

3. If ω is imaginary cube root of unity, then $\omega+\omega^{1+2\left(\frac{3}{4}\right)+3\left(\frac{3}{4}\right)^2+4\left(\frac{3}{4}\right)^3+\ldots\ldots}$ is:

(A) 2ω **(B)** -1 **(C)** $1+\omega$ **(D)** None of these

4. The range of values of 'a' such that angle θ between pair of tangents drawn from $(a, 0)$ to the circle $x^2+y^2=1$ lies in the interval $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ is:

(A) $\left(-\sqrt{3}-\sqrt{2}, -2\right)\cup\left(2, \sqrt{3}+\sqrt{2}\right)$ **(B)** $\left(-2, -\sqrt{2}\right)\cup\left(\sqrt{2}, 2\right)$

(C) $\left(-\sqrt{6}-\sqrt{2}, -2\right)\cup\left(2, \sqrt{6}+\sqrt{2}\right)$ **(D)** None of these

5. Consider a parabola $y^2=8x$ & an ellipse $\frac{x^2}{4}+\frac{y^2}{15}=1$, let L be the common tangent touching parabola at P & ellipse at Q, then length of PQ is:

(A) $\frac{17}{4}\sqrt{3}$ **(B)** $\frac{17}{4}\sqrt{2}$ **(C)** $\frac{17}{2}$ **(D)** $\frac{17}{4}\sqrt{5}$

6. If the chord of contact of tangents from a point P to the parabola $y^2=4ax$ touches the parabola $x^2=4by$, then locus of P is:

(A) $by^2=4a^2x$ **(B)** $xy=2ab$ **(C)** $xy=-2ab$ **(D)** $xy=-4a^2b$

SPACE FOR ROUGH WORK

SECTION 2
MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. PQ is the latus rectum of the parabola $y^2 = 4x$, ΔPQR is isosceles right angled triangle, where R is on the axis of the parabola, line PR meets the parabola at S & QR meet the parabola at T, then which of the following is/are TRUE?

(A) the area of trapezium $PQST$ is 64 sq.units

(B) the circum radius of trapezium $PQST$ is $2\sqrt{10}$ units

(C) the area of trapezium $PQST$ is 72 sq.units

(D) the circum radius of trapezium $PQST$ is $8\sqrt{2}$ units

8. Consider the equation $z^2 - (3+i)z + m + 2i = 0,\ m \in R$.

If the equation has exactly one real α & one non real root β, then which of the following can be TRUE?

(A) Modulus of non-real imaginary root is 2

(B) $m = 3$

(C) $\left(\dfrac{\beta}{\bar{\beta}}\right)^{\frac{1}{\alpha}} = \dfrac{\beta}{\sqrt{\alpha}}$

(D) $\left(\dfrac{\bar{\beta}}{\beta}\right)^{\frac{1}{\alpha}} = -\dfrac{\bar{\beta}}{\sqrt{\alpha}}$

9. Let PA & PB are two tangents drawn from point P, to the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1,\ (a > b)$. If T_1 be reflection of focus S_1, about tangent PA & T_2 be reflection of focus S_2 about tangent PB, then:

(A) $\angle PT_2S_1 = \angle PS_2T_1$

(B) mid-point of line joining S, T_1 lies on circle $x^2 + y^2 = a^2$

(C) ΔPS_2T_1 & ΔPT_2S_1 are similar

(D) ΔPS_2T_1 & ΔPT_2S_1 are congruent

10. Locus of the centre of a circle touching the circle $x^2 + y^2 - 2x - 4y - \dfrac{7}{4} = 0$ internally & tangents on which from (1, 2) is making an angle 60º with each other is director circle C of a variable ellipse (whose axis are parallel to coordinate axis), then which of the following is(are) true?

(A) Director circle of C is $(x-1)^2 + (y-2)^2 = 6$

(B) Director circle of C is $(x-1)^2 + (y-2)^2 = 3$

(C) Equation of ellipse can be $(x-1)^2 + \dfrac{(y-2)^2}{2} = 1$

(D) Equation of ellipse can be $\dfrac{(x-1)^2}{2} + (y-2)^2 = 1$

11. If z be a complex number such that $\frac{z-4}{z-2i}$ is purely imaginary, then $w=\frac{z}{2}$ satisfies:

(A) $\frac{w-2}{w-i}=ik,\ k\in R-\{0\}$ **(B)** $\frac{w}{w-2-i}=ik,\ k\in R-\{0\}$

(C) $\frac{w-2}{2w-i}=k,\ k\in R-\{0\}$ **(D)** $\left|w-1-\frac{1}{2}i\right|=\frac{\sqrt{5}}{2}$

12. The curve described parametrically by $x=t^2+t+1,\ y=t^2-t+1$, has:

(A) Vertex at (1, 1) **(B)** Vertex at (0, 0)

(C) Latus rectum of length $\sqrt{2}$ **(D)** Latus rectum of length 2

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SECTION 3
NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

13. Minimum distance between the curves $y^2=4x$ & $x^2+y^2-12x+31=0$ is ________.

14. If length of chord of contact from origin to the circle $x^2+y^2-4ax-2ay+4a^2=0,\ (a>0)$ is 3, then value of $\frac{\sqrt{5}}{a}$ is ________.

15. If $|z_1|=1, |z_2|=\sqrt{3}, |z_3|=\sqrt{5}, |z_4|=\sqrt{7}$, then $\left(\frac{|105z_1+35z_2+21z_3+15z_4|}{|z_2z_3z_4+z_1z_3z_4+z_1z_2z_4+z_1z_2z_3|}\right)^2$ is_______ .

16. If the mid-point of a chord of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is $\left(2,\frac{5}{2}\right)$, then length of the chord is:

17. If the tangents are drawn from any point on the line $x+y=3$ to the circle $x^2+y^2=9$. Then all possible chords of contact pass through a fixed point (h, k), then $h+k$ is ______.

18. If $y=2x+c$ is a normal to the parabola $y^2=4x$, then $|c|$ is ______.

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